

WASP-43b

R-0501

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_230309 · created 2026-07-20T16:08:28+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460012.74117 +/- 0.00098 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1981 +/- 0.0098
Transit depth (Rp/Rs)^2	3.92 +/- 0.39 [%]
Orbital Inclination (inc)	82.29 +/- 1.12
Airmass coefficient 1 (a1)	1.019 +/- 0.016
Airmass coefficient 2 (a2)	-0.011 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	#3 - [564, 176]
Optimal Method	PSF photometry
Transit Duration (day)	0.0539 +/- 0.0033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1981 ± 0.0098 vs 0.1594 (deviation 3.87σ)
Duration fit vs published	0.0539 d vs 0.0512 d (deviation 0.79σ)
Mid-time O–C	4.9 min
Depth SNR	19.8
Residual scatter	1.53 %

Light curve

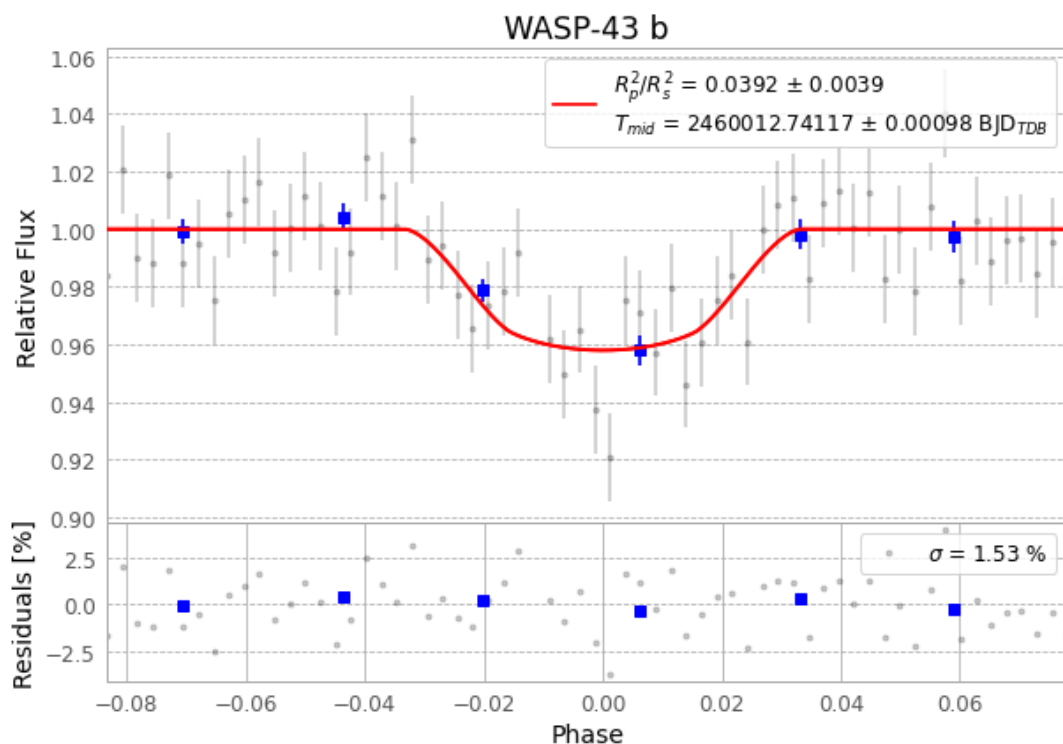


Figure 1 — FinalLightCurve_WASP-43 b_2023-03-08.png (SHA-256 03cf113f2e6c940a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [442, 89], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 367da60ca4845310...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

64 FITS frames (42 MB), WASP-43230309040015.FITS → WASP-43230309070915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-230309.log (7a70c2447832...), FinalLightCurve_WASP-43 b_2023-03-08.png (03cf113f2e6c...), FinalLightCurve_WASP-43 b_2023-03-08.csv (4bd8264f7aca...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 16:08Z.